

ENTERPRISE AI CAPITAL ALLOCATION SERIES — PART 2 OF 2

Funding the Shift to Autonomous AI

A 100-Agent Fleet Cost Model and Architecture Playbook (2027–2028) — builds on the Part 1 disclosure analysis of 15 non-hyperscaler enterprises.

Critical framing — read this before the numbers

Everything in this document is a **modeled, illustrative estimate**, built bottom-up from real, currently published 2026 inference pricing, tooling pricing, and documented case studies of agent cost behavior. **None of it is company-disclosed data.** No company in the Part 1 list — or anywhere else — publicly discloses a "100-agent fleet" cost line. Where Part 1 said "not disclosed separately," this document does not fill that gap with a disguised estimate; it builds a separate, clearly-labeled model and shows its assumptions and sources so you can stress-test or replace any input.

Two things make this modeling exercise unusually volatile compared to normal IT cost forecasting:

1. **Per-token prices are falling** — Claude Opus dropped 67% (\$15/\$75 → \$5/\$25 per million input/output tokens) between Opus 4.1 (mid-2025) and Opus 4.6 (Feb 2026); OpenAI's GPT-5.4 sits at \$2.50/\$15.
2. **Token consumption is rising much faster than prices are falling.** Goldman Sachs Research projects enterprise/consumer token consumption will multiply **24x between 2026 and 2030** (to 120 quadrillion tokens/month), driven specifically by the shift from single-turn queries to "always-on" background agents.

That combination — cheaper unit price, exploding volume — is exactly why real enterprises are already getting burned: Uber exhausted its entire 2026 AI coding budget in four months after rolling Claude Code out to ~5,000 engineers; one unnamed enterprise reportedly spent **\$500 million in a single month** after deploying AI access with no usage caps; a documented 100-agent coding fleet (Peter Steinberger's OpenClaw project, May 2026) generated **603 billion tokens and \$1.3 million in spend over 30 days** — annualized, roughly **\$15.6M/year for 100 agents** — with 70% of that traceable to a single unmanaged setting (a high-rate "Fast Mode" configuration).

That \$15.6M/year figure is the cautionary upper bound this model deliberately does **not** assume. It is real, sourced, and should sit in the back of your mind as "what happens with zero governance." The two archetypes below assume competent FinOps discipline at different levels of model-tier ambition.

Deliverable 3: Illustrative 2027–2028 Cost Model for a 100-Agent, 24/7 Fleet

Stated assumptions

Change any of these and the output changes proportionally.

Assumption	Value used	Source basis
Fleet size	100 agents, running continuously (24/7/365)	Per stated scope
Cycle cadence	288 reasoning cycles/agent/day (~1 every 5 min)	Illustrative; consistent with "always-on background agent" framing in Goldman Sachs Research commentary
Cycle mix	Lightweight "heartbeat" vs. heavier "active task" cycles — ratio differs by archetype	Reflects the cost-accumulation pattern documented in LeanOps' 2026 audit of 30 production agentic teams
Context growth	Active cycles carry larger accumulated context (10K–30K input tokens) than heartbeat cycles (2K–5K)	LLM APIs are stateless — agents resend full history each step. LeanOps 2026 audit; TechAhead 2026 analysis
Model pricing	Haiku 4.5 \$1/\$5, Sonnet 4.6 \$3/\$15, Opus 4.8 \$5/\$25 per MTok; GPT-5.4 \$2.50/\$15, GPT-5.5 \$5/\$30	Anthropic and OpenAI published rate cards, May–June 2026
Prompt caching	Cached input reads run ~90% cheaper than fresh input on Claude models	Anthropic pricing documentation
Batch API	Not modeled in the base case — flagged as an available lever, not baked in	Would add a further 50% discount on tolerant workloads

Archetype A — "Pure OpEx Reallocator"

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Capital One / Block pattern: existing cloud commitment, aggressive model routing and caching, budget reallocated from existing tech spend rather than net-new.

PER-AGENT DAILY PROFILE (MODELED)

230 heartbeat cycles/day at 2,000 input / 200 output tokens, routed to Haiku 4.5, 80% cache-hit rate on input; 58 active-task cycles/day at 15,000 input / 1,000 output tokens, routed to Sonnet 4.6, 50% cache-hit rate.

Cost component	2027 (modeled)	2028 (modeled)	Basis / caveat
Inference/token cost	~\$97,000/yr	~\$113,000/yr	Bottom-up calc; 2028 assumes ~45% volume growth partly offset by ~20% further price decline
FinOps/observability tooling	~\$45,000/yr	~\$50,000/yr	Self-hosted Langfuse + hosting/maintenance
Data infra (context/memory)	~\$35,000/yr	~\$40,000/yr	Incremental storage layered onto existing cloud commitment
Orchestration overhead	~\$75,000/yr	~\$85,000/yr	Open-source frameworks — mostly engineering labor, not licensing
Annual total	~\$252,000	~\$288,000	

2-year total (2027–2028, modeled): ~\$540,000

Archetype B — "High-Inference Scale"

JPMorgan / Bank of America pattern: frontier-model tolerance, net-new dedicated compute budget, larger context windows, lower caching discipline than Archetype A.

PER-AGENT DAILY PROFILE (MODELED)

172.8 heartbeat cycles/day at 5,000 input / 500 output tokens, routed to Sonnet 4.6, 20% cache-hit rate; 115.2 active-task cycles/day at 30,000 input / 2,000 output tokens, routed to Opus 4.8, 20% cache-hit rate.

Cost component	2027 (modeled)	2028 (modeled)	Basis / caveat
Inference/token cost	~\$852,000/yr	~\$988,000/yr	Same growth logic as Archetype A, applied to frontier-model, lower-caching profile
FinOps/observability tooling	~\$275,000/yr	~\$320,000/yr	Enterprise-tier managed observability + a dedicated agentic spend-governance layer
Data infra (context/memory)	~\$175,000/yr	~\$210,000/yr	Dedicated vector DB/context-persistence infrastructure
Orchestration overhead	~\$225,000/yr	~\$260,000/yr	Managed agent-runtime infrastructure, redundancy/failover
Annual total	~\$1,527,000	~\$1,778,000	

2-year total (2027–2028, modeled): ~\$3,305,000

Sanity-check against real-world data points

These modeled figures sit **well below** Steinberger's documented \$15.6M/year run-rate for 100 coding agents in an unmanaged, Fast-Mode-heavy configuration — intentionally, since that case is the "no governance" outcome, not a target architecture. They also sit in a plausible range relative to LeanOps' documented 20-engineer agentic coding team, whose *unoptimized* monthly bill was \$87,000 (\$1.04M/year) before FinOps intervention brought it to \$24,000/month (\$288K/year) — a different workload shape but directionally consistent with the roughly 4x optimization delta this model implies between disciplined and undisciplined operation.

DO NOT OVER-READ THE PRECISION

Treat the dollar figures above as directional, not forecasts. Treat the *ratio* between Archetype A and B (roughly 6x), and the *direction* of year-over-year change, as the more defensible takeaways than any single number.

Deliverable 4: Architectural Takeaways for a 100-Agent Fleet

For the Pure OpEx Reallocator track — design for strict token discipline

Model cascading/routing

Route by default to the cheapest model capable of the task; escalate only on demonstrated need. The documented pattern is a three-tier cascade — cheap/fast model (Haiku-class) for classification and routing → mid-tier model (Sonnet-class) for general reasoning → frontier model (Opus-class) reserved for the hardest agentic or long-context work. This is described as "the single biggest cost lever" available before any other optimization.

Prompt/context caching as a default

Cached input reads run at roughly 10% of fresh-input price on current Claude pricing. Structuring agent prompts so stable system instructions sit in a cacheable prefix is described as "a 10x cost reduction with no engineering work" when done correctly — the highest-leverage, lowest-effort lever in this entire model.

Batch API for tolerant workloads

Any workload that doesn't need sub-second response — nightly reconciliation, batch classification, non-urgent summarization — should route through Batch APIs for a flat 50% discount on both input and output, stackable with caching.

Sliding-window context pruning

One documented remediation (the LeanOps case study) added "context pruning with a sliding window of the last 8 steps" and captured a 12% cost reduction on its own, on top of model-routing savings.

Hard per-agent and per-fleet budget caps, enforced at the gateway

The documented remediation pattern is consistent across multiple 2026 case studies: a soft daily cap with an alert, a hard daily cutoff that forces a downgrade to the cheapest model, and a monthly ceiling that requires manual approval to exceed. One practitioner reports this pattern "catches 95% of runaway patterns before they happen."

For the High-Inference Scale track — design for loop boundaries and enterprise FinOps

Loop boundaries with machine-verifiable exit criteria

The most robust pattern for long-running autonomous work (the "Ralph Loop" architecture and similar) starts each iteration with a *fresh* context window, reads current state from persistent storage, does bounded work, writes updated state back, and repeats — with an explicit, machine-checkable completion criterion rather than an open-ended "keep going" instruction.

Spawn budgets and iteration ceilings

Any agent capable of spawning sub-agents or retrying failed steps needs an explicit cap on total spawns/retries per task. This is the documented fix for the "recursive loop" failure mode that produced the DN42 incident's \$6,531 runaway AWS bill, and is architecturally identical to the failure that drove Uber past its full annual AI budget in four months.

Circuit breakers on token velocity, not just total spend

A rate-based control (tokens-per-minute) catches a runaway loop in real time — one documented implementation caught a runaway loop within 60 seconds by tripping at a 10,000 tokens/minute threshold, versus a healthy agent's normal sustained rate of 3,000–4,000 tokens/minute.

Explicit state persistence

For 100 agents running continuously for two years, state cannot live only in-context. Current architectural guidance converges on the same principle: externalize state to durable storage, checkpoint at defined boundaries, and design for graceful recovery after interruption.

Enterprise FinOps as a control plane, not a dashboard

Observability tools tell you what happened after the fact; only a small, newer category of tooling actively *enforces* budgets in real time. The FinOps Foundation's own 2026 commentary is blunt: the industry conversation shifted in months from "move fast" to "we need guardrails" — after multiple enterprises found themselves several multiples over their entire annual token budget by April. Build the enforcement layer before the fleet goes live, not after the first surprise invoice.

Consolidated Caveats

- **Modeled, not disclosed.** All dollar figures in Deliverable 3 are modeled outputs of stated assumptions. Treat every number as replaceable — the value here is the *structure* of the model, not the specific dollars.
- **Pricing is a moving target.** Current as of mid-2026 search dates. Anthropic and OpenAI have both changed per-token pricing multiple times within the past 12 months. Re-verify rate cards before using this model for an actual budget submission.
- **Archetypes are illustrative.** "High-Inference Scale" and "Pure OpEx Reallocator" are inspired by the spending patterns in Part 1, not a claim that those specific companies run agent fleets shaped exactly like this model.
- **Outcomes are bimodal, not normal.** Real-world results split between well-governed (LeanOps' optimized \$24K/month case) and ungoverned (Steinberger's \$1.3M/month case), with limited middle ground reported so far. Budget for the governance work, not just the inference.